

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

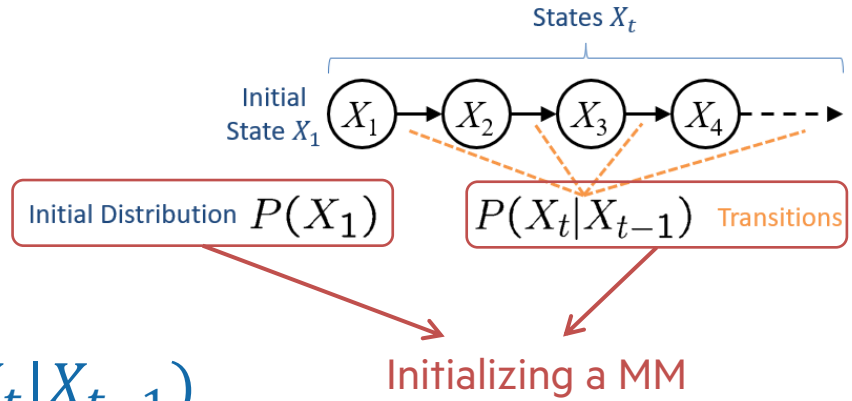
- The grades of Homework 1 and Classwork 4 are available on Brightspace.
  - HW1 Friendly Competition Scores
    - 1st Place: -356
    - 2nd Place: - 358
    - 3rd Place: -362
- Homework 2
  - Due this **Friday 10/25 @5 pm**

This Lecture

- Hidden Markov Model (HMM)
  - A more general format of Markov Models (MMs)
- Filtering
  - Forward Algorithm

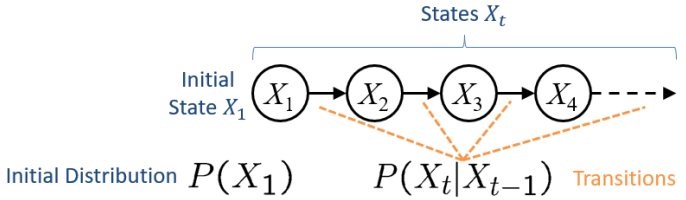
Last Lecture

- Markov Models
  - **State** Variable: X_t
    - Initial Distribution: $P(X_1)$
  - **Transition** Probabilities: $P(X_t|X_{t-1})$
  - What's the **goal** of MM?
    - Predicting **future** states <u>only</u> on the **current** state
  - What's the **Markov property**?
    - **future** ⫫ **past** |present
    - It is a chain and d-separation applies



Why?

Last Lecture



- Joint Distribution Factorization

$$P(X_1, X_2, \dots, X_T) = P(X_1) \prod_{t=2}^{T} P(X_t|X_{t-1})$$

- Mini-Forward Algorithm
  - Arbitrary $P(X_t)$
  - $P(X_t = x_t) = \sum_{x_{t-1} \in X_{t-1}} P(X_t = x_t | X_{t-1} = x_{t-1}) P(X_{t-1} = x_{t-1})$

All possibilities of previous state

Example – Sentence Generation

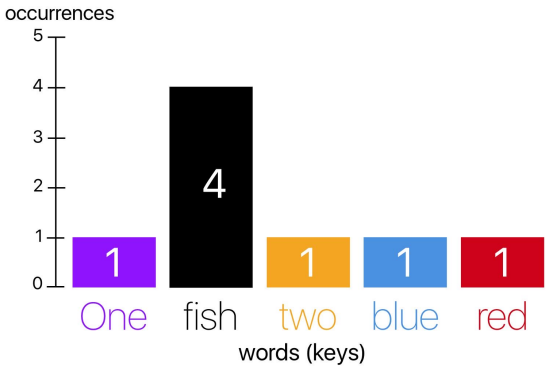
- Starter Sentence

  One fish two fish red fish blue fish

- Terminologies
  - Token: word
  - Key: unique word

How many **tokens** and **keys** in the starter sentence?

One : 1
fish : 4
two : 1
red : 1
blue : 1



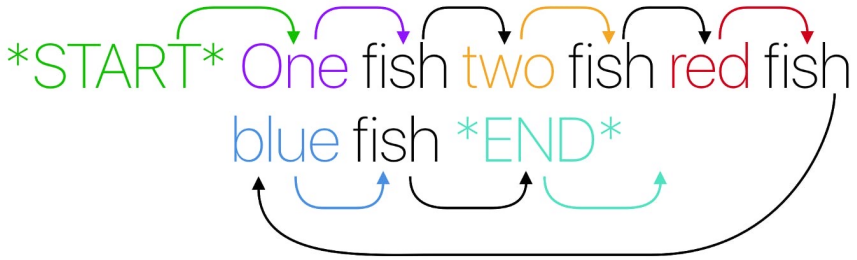
Source: https://hackernoon.com/from-what-is-a-markov-model-to-here-is-how-markov-models-work-1ac5f4629b71

Example – Sentence Generation

- Adding two unique tokens: “start” and “end”

START One fish two fish red fish
blue fish *END*

- Markov Model

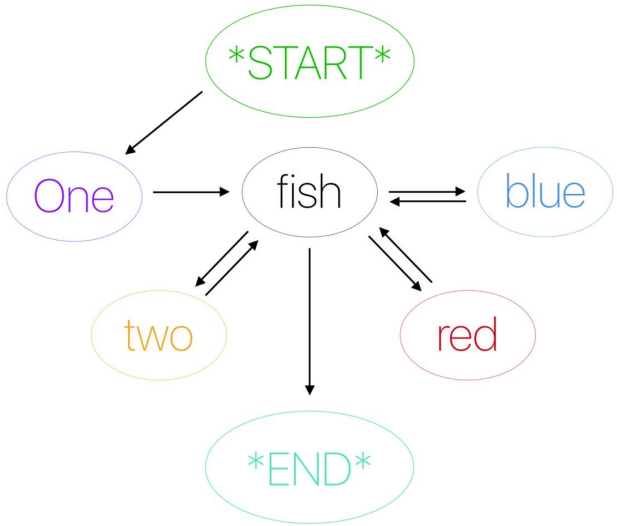


(X_{t-1}, X_t)

(*Start*, One)
(One, fish)
(fish, two)
(two, fish)
(fish, red)
(red, fish)
(fish, blue)
(blue, fish)
(fish, *END*)
(*END*, none)

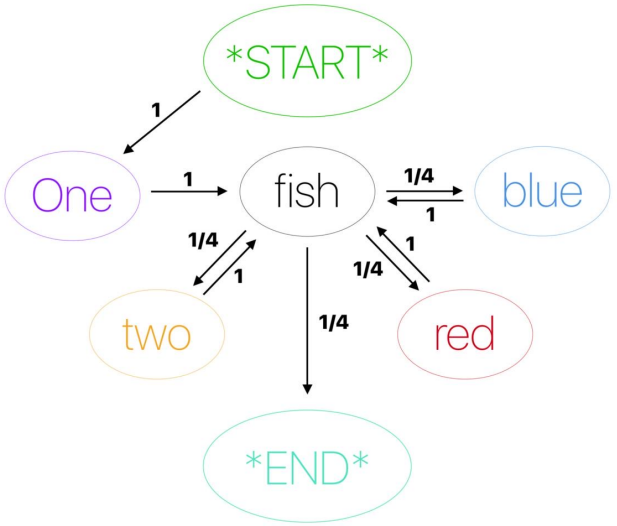
Example – Sentence Generation

What is missing?



One fish two fish red fish blue fish

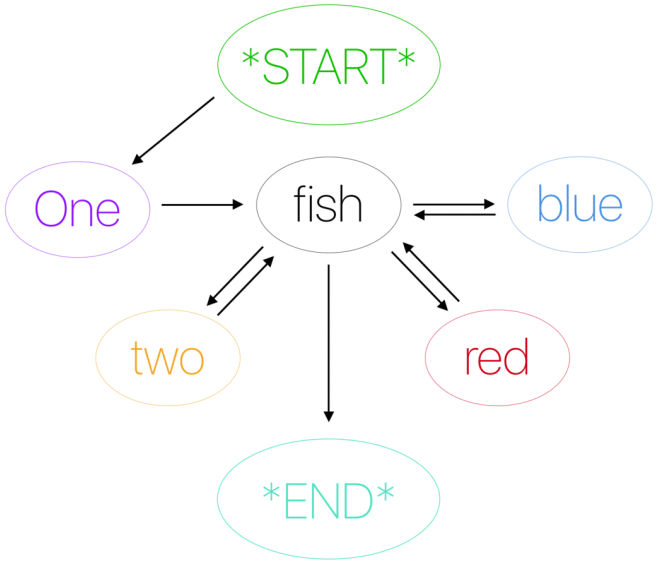
Transition Probabilities



One fish two fish red fish blue fish

Example – Sentence Generation

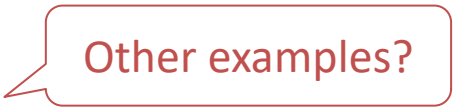
- Finally!



GIF: https://hackernoon.imgix.net/hn-images/1*MbHRwYNA8F29hzes8EPHiQ.gif
An implementation of MM for text generation: https://github.com/jsvine/markovify

Limitation of Markov Model

- The state variables often cannot be directly observed.
  - Weather Prediction: {sunny, rainy, cloudy} 
  - Car Engine Health
    - State: {healthy, mild wear, severe wear}?
    - Observable:
      - Vibration patterns
      - Engine noise levels
      - Fuel efficiency

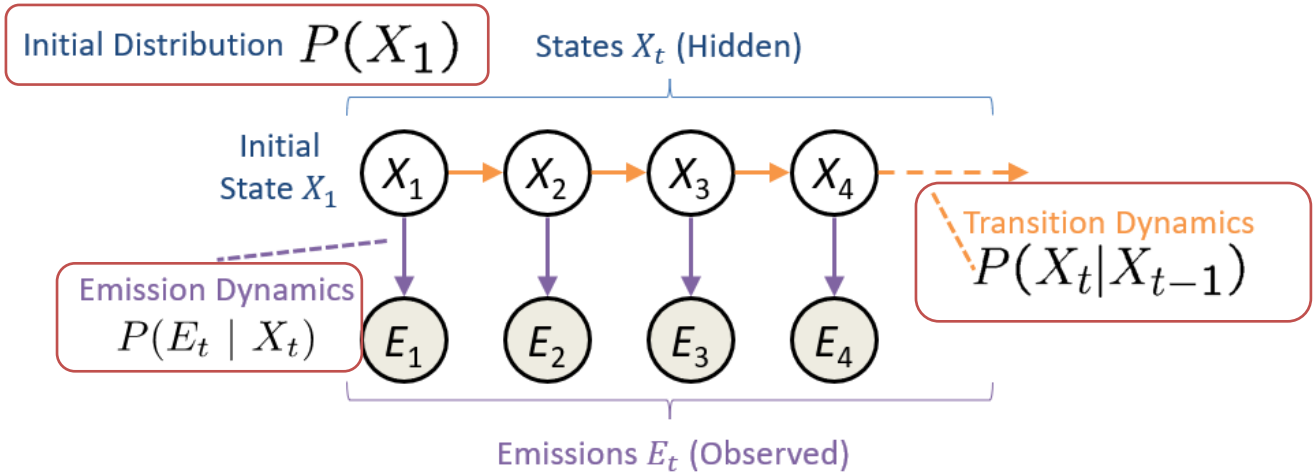


Hidden Markov Model

- **State** is still changing over time/space, but its direct observation is not possible for the agent (hidden).
- **Emissions** are observations that emanate **from the current state.**
- Goal: use all **observed emissions** up until the **current time *t*** to make assessments about the **next state**.

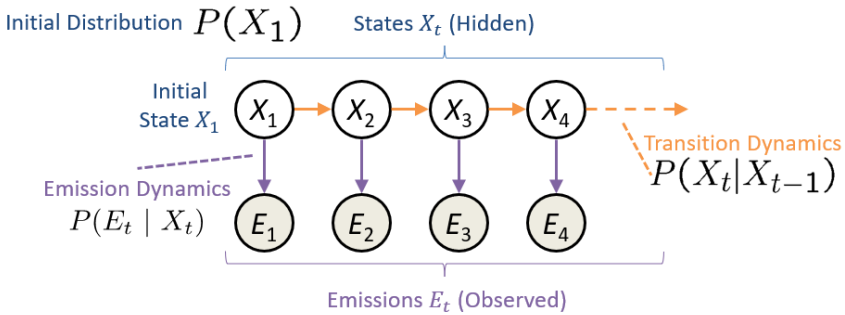
Hidden Markov Model

- A **Hidden Markov Model** (HMM) assumes:
  - An underlying **MM** is **not observed**
  - **Emissions** from the underlying MM are **observed**.



Joint Distribution Factorization in HMM

- Can you factor $P(X_1, E_1, X_2, E_2, \ldots, X_T, E_T)$?



$$P(X_1, E_1, X_2, E_2, \ldots, X_T, E_T) = P(X_1)P(E_1|X_1)\prod_{t=2}^{T} P(X_t|X_{t-1})P(E_t|X_t)$$

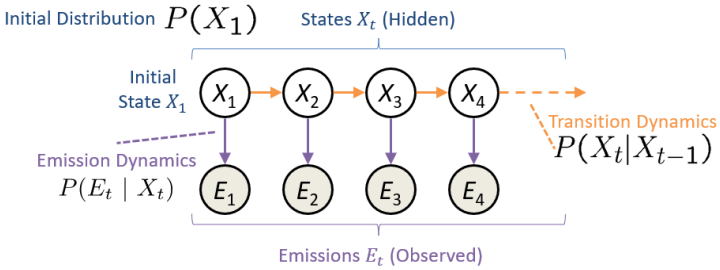
Mini-Forward Algorithm in MM

- In MM, to compute arbitrary $P(X_t)$:

$$P(X_t = x_t) = \sum_{x_{t-1} \in X_{t-1}} P(X_t = x_t | X_{t-1} = x_{t-1}) P(X_{t-1} = x_{t-1})$$

- Can we still apply this in HMM?
  - No
    - States are **hidden** and we only have access to emissions
    - Cannot propagate state probabilities forward because we **don't know the true states**

Filtering



- Belief State $B(X_t)$
  - The likelihood of the state X_t given all past evidence
  - Formally, $B(X_t) = P(X_t|E_1 = e_1, E_2 = e_2, \dots, E_t = e_t)$
    $= P(X_t|e_{1:t})$
- Filtering
  - The process of tracking and updating $B(X_t)$
- Inference Goal of HMM: $B(X_t) = P(X_t|e_{1:t})$

Forward Algorithm

- Answering an arbitrary $B(X_t)$

$$B(X_t = x_t) = P(x_t|e_{1:t})$$

$$= \frac{P(x_t, e_{1:t})}{P(e_{1:t})}$$

Conditioning

$$P(x_t, e_{1:t}) = \sum_{x_{t-1}} P(e_{1:t}, x_t, x_{t-1})$$

Law of Total Prob.

$$= \sum_{x_{t-1}} P(e_t, x_t, x_{t-1}, e_{1:t-1})$$

Break $e_{1:t}$

Forward Algorithm

$$P(x_t, e_{1:t}) = \sum_{x_{t-1}} P(e_t, x_t, x_{t-1}, e_{1:t-1})$$

Conditioning: P(A,B)= P(A|B)P(B)

$$= \sum_{x_{t-1}} P(e_t | x_t, x_{t-1}, e_{1:t-1}) P(x_t, x_{t-1}, e_{1:t-1})$$

$$= \sum_{x_{t-1}} P(e_t | x_t, x_{t-1}, e_{1:t-1}) P(x_t | x_{t-1}, e_{1:t-1}) P(x_{t-1}, e_{1:t-1})$$

$$= \sum_{x_{t-1}} P(e_t | x_t) P(x_t | x_{t-1}) P(x_{t-1}, e_{1:t-1})$$

Independence of HMM

$$= P(e_t | x_t) \sum_{x_{t-1}} P(x_t | x_{t-1}) P(x_{t-1}, e_{1:t-1})$$

Rearrange

Forward Algorithm

$$B(X_t = x_t) = P(x_t|e_{1:t})$$

$$\propto P(x_t, e_{1:t})$$

$$= P(e_t|x_t) \sum_{x_{t-1}} P(x_t|x_{t-1}) P(x_{t-1}, e_{1:t-1})$$

$$= P(e_t|x_t) \sum_{x_{t-1}} P(x_t|x_{t-1}) \cdot \alpha B(x_{t-1})$$

P(A|B) = P(A,B)/P(B)

- $\propto$ P(A,B)
- = α P(A,B)

α is the normalizing constant

emission dynamics

transition dynamics

belief of being in last state

Forward Algorithm: "Push" belief states from earlier time states **forward** -- if you knew $B(X_{t-1})$, we could use that as a "checkpoint" through which to compute $B(X_t)$.